

7.1 Waves: Transverse & Longitudinal

Question Paper

Course	CIEA Level Physics
Section	7. Waves
Topic	7.1 Waves: Transverse & Longitudinal
Difficulty	Medium

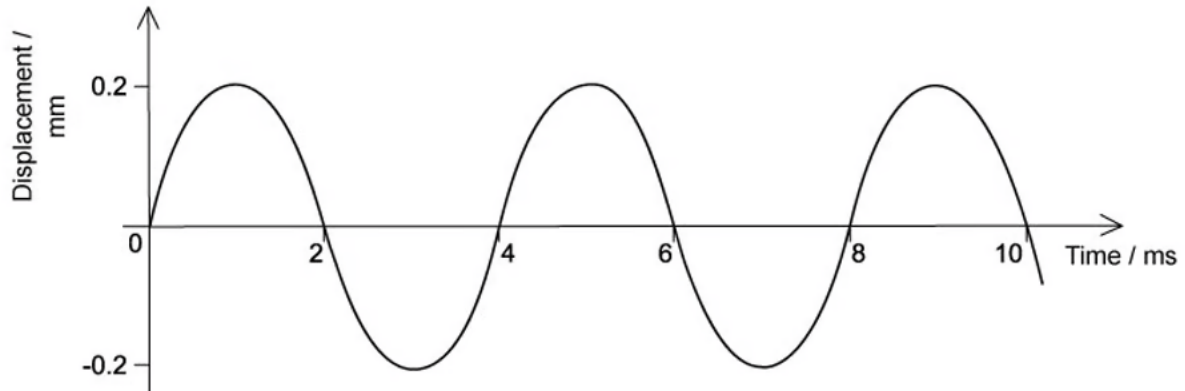
Time allowed: 20

Score: /11

Percentage: /100

Question 1

Sound travels with a speed of 330 m s^{-1} in air. The variation with time of the displacement of an air particle due to a sound wave is shown below.



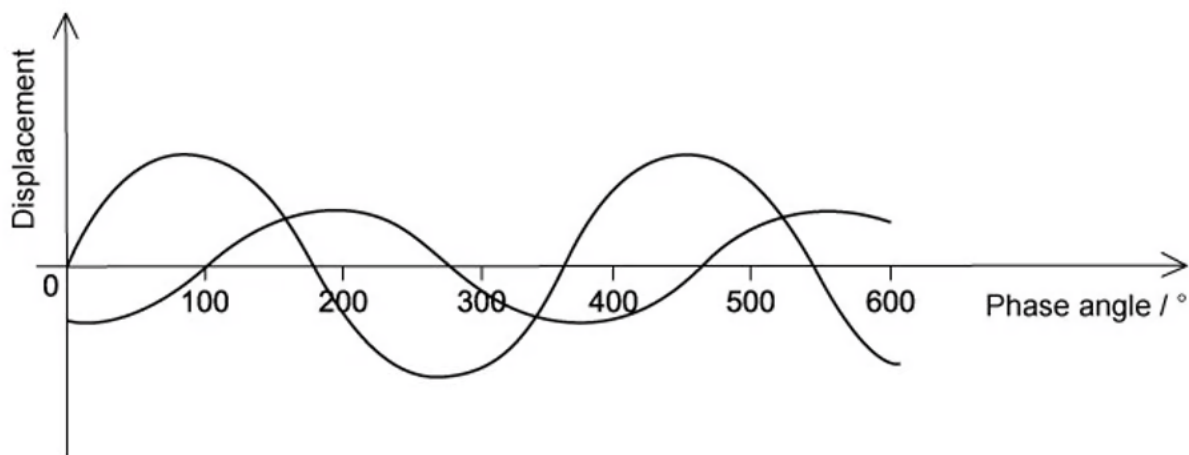
Which of the following statements about the wave is correct?

- A. the wavelength of the sound wave is 1.32 m
- B. the graph shows that the sound is a transverse wave
- C. the frequency of the wave is 500 Hz
- D. the intensity of the wave will be doubled if its amplitude is increased to 0.4 mm

[1 mark]

Question 2

The graph shows two light waves travelling at the same frequency.



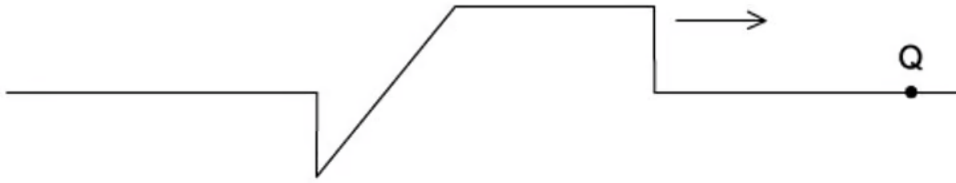
Which of the following could be the phase difference between the two waves?

- A. 330°
- B. 260°
- C. 220°
- D. 150°

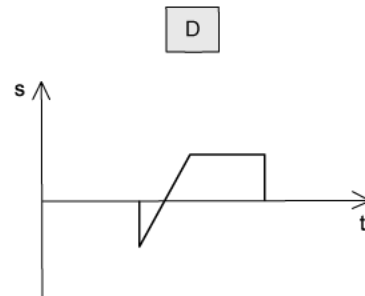
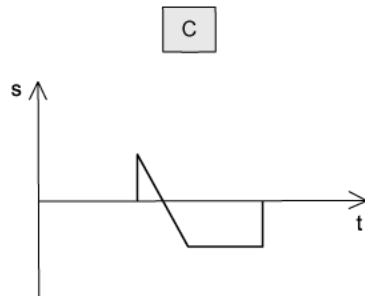
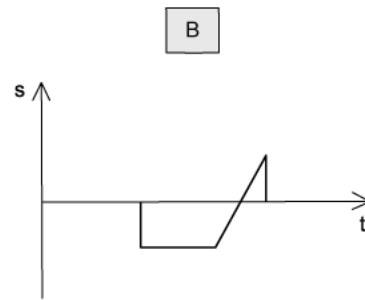
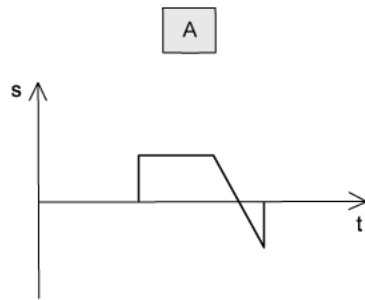
[1 mark]

Question 3

The diagram shows a wave moving along a stretched rope in the direction shown.



Which of the following will be the correct graph to show the variation with time t of the displacement s of the particle Q in the rope?



[1 mark]

Question 4

A wave is travelling through a medium at 24 m s^{-1} with a frequency of 15 Hz .

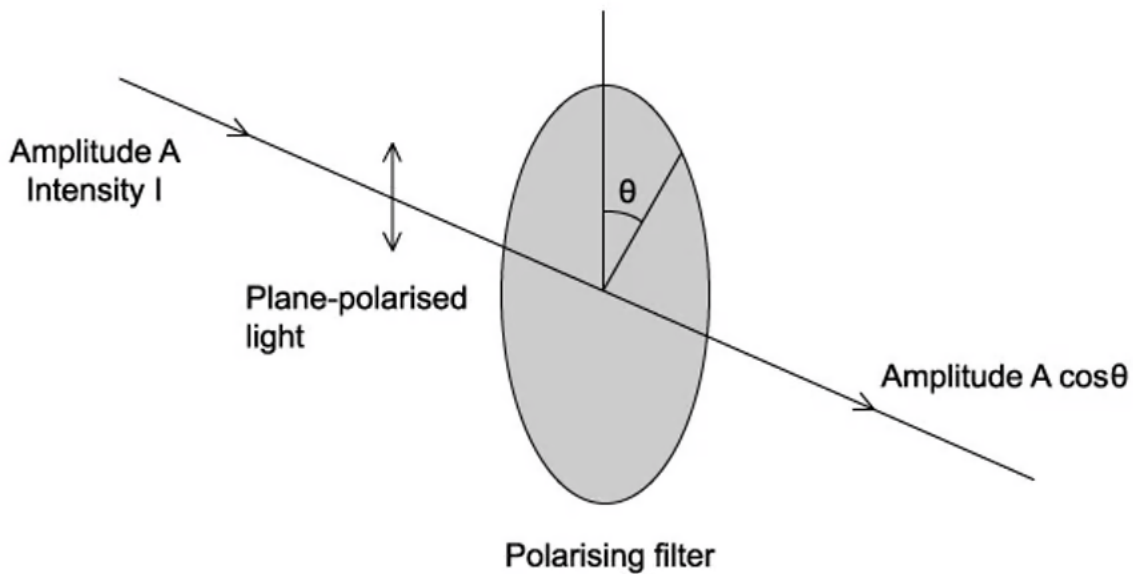
What is the phase difference between two points 2 m apart?

- A. They are out of phase by half a cycle.
- B. There is no phase difference.
- C. They are out of phase by a quarter of a cycle.
- D. They are out of phase by 0.8 of a cycle.

[1 mark]

Question 5

A plane polarised light of amplitude A is passed through a polarising filter as shown in the diagram. The emerging light has an amplitude of $A \cos \theta$



If the intensity of the beam is I , what is the intensity of the emerging light when θ is 60.0° ?

- A. $0.866 I$
- B. $0.750 I$
- C. $0.500 I$
- D. $0.250 I$

[1 mark]

Question 6

A police car siren emits a sound wave with a frequency f_s of 440 Hz. The car is travelling away from an observer at a speed of 30 m s^{-1} .

The speed of sound is 340 m s^{-1} .

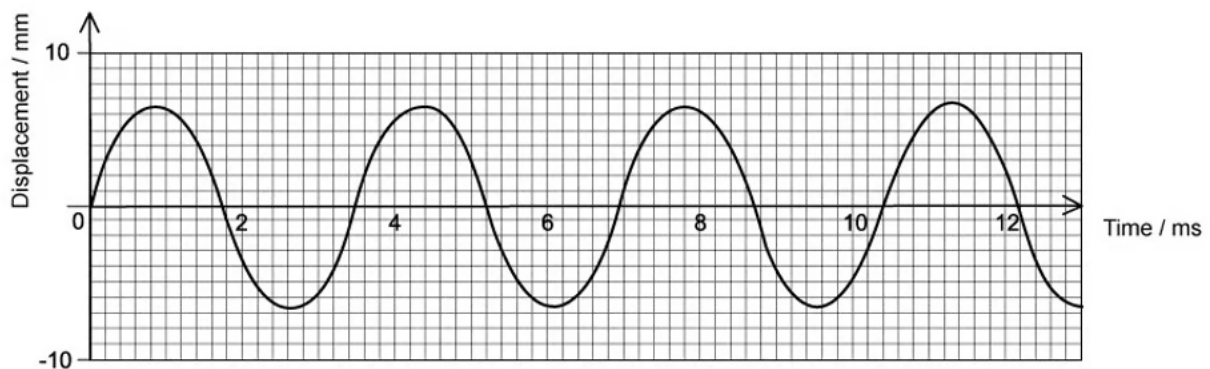
Which of the following is the frequency the observer hears?

- A. 478 Hz
- B. 460 Hz
- C. 440 Hz
- D. 404 Hz

[1 mark]

Question 7

What is the frequency, amplitude and period of the wave shown in the graph below, to 2 significant figures?



	Period / s	Frequency / Hz	Amplitude / m
A	0.0042	240	0.013
B	0.0035	290	0.0065
C	0.0031	320	0.013
D	0.0027	290	0.0065

[1 mark]

Question 8

A sound wave with an amplitude a of 0.2 mm has an intensity of 2.5 W m^{-2}

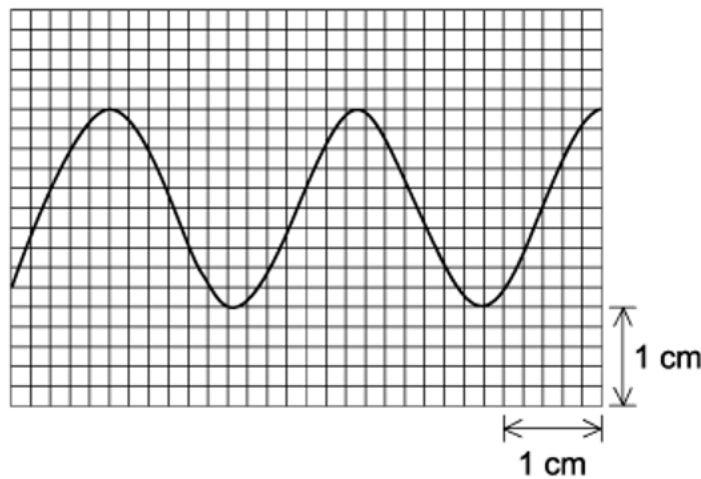
If a wave with the same frequency has an amplitude of $2a$ what is the intensity of the wave?

- A. 10.0 W m^{-2}
- B. 9.0 W m^{-2}
- C. 6.0 W m^{-2}
- D. 4.2 W m^{-2}

[1 mark]

Question 9

A sound wave can be displayed on a cathode ray oscilloscope (C.R.O.). The time base is set to $5 \mu\text{s mm}^{-1}$.



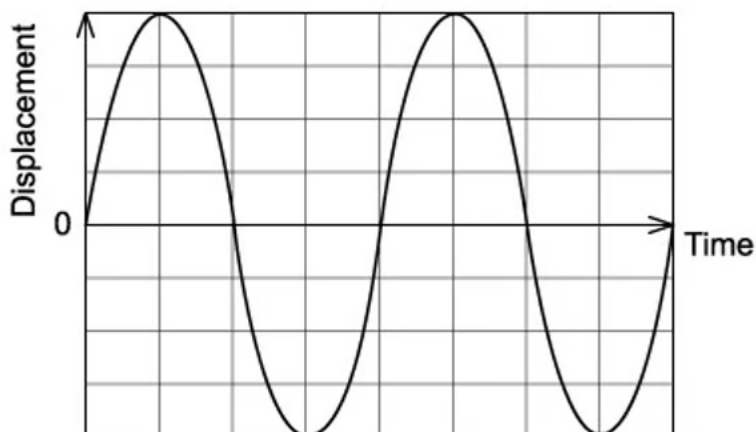
What is the frequency of the sound wave?

- A. 83 Hz
- B. 8.3 Hz
- C. 83 kHz
- D. 8.3 kHz

[1 mark]

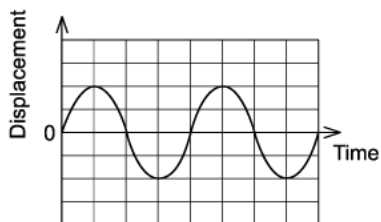
Question 10

The graph shows the displacement of a sound wave over time.

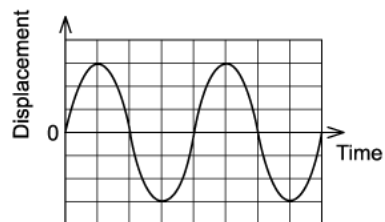


When the intensity of the wave is halved, which graph shows the correct displacement of the wave?

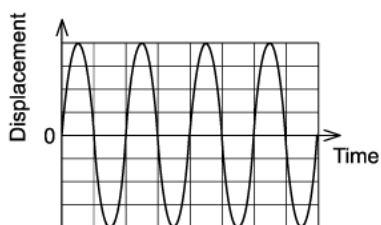
A



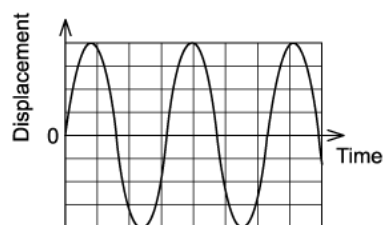
B



C



D



[1 mark]

Question 11

A source of sound power P is situated in an open space. The intensity I of sound at distance r from this source is given by

$$I = \frac{P}{4\pi r^2}$$

How does the amplitude A of the vibrating air molecules vary with the distance r from the source?

A. $A \propto \frac{1}{r}$

B. $A \propto \frac{1}{r^2}$

C. $A \propto r$

D. $A \propto r^2$

[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- The diagram shows a displacement time graph of a sound wave which is a longitudinal wave
- As time is on the x-axis, we can work out the period but not the wavelength of the wave
- Two wavelengths have a period of $2T = 8 \text{ ms}$
 - So, $T = 4 \text{ ms}$
- From this frequency can be calculated from the equation
 - $f = \frac{1}{T} = \frac{1}{4 \times 10^{-3}} = 250 \text{ Hz}$
- The speed of the wave is calculated by the equation $v = f \lambda$
- Re-arrange for wavelength, $\lambda = \frac{330}{250} = 1.32 \text{ m}$

B is incorrect because the graph shows a sine wave or sinusoid wave, this is a graphical depiction of the wave that describes a periodic oscillation this will be seen for both transverse and longitudinal waves

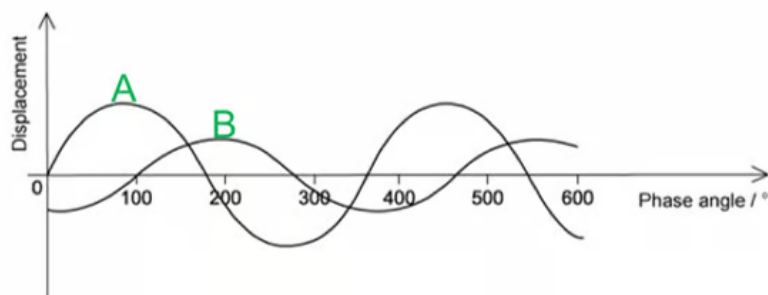
C is incorrect because when the frequency is calculated from the graph it is 250 Hz, not 500 Hz

D is incorrect because the intensity of the wave depends on the square of the amplitude. Doubling of the amplitude multiplies the intensity by a factor of 4

2

The correct answer is **B** because:

- The question states that both waves have the same frequency
- If the wave with the larger amplitude is labelled A and the lower amplitude B



- As the wave is sinusoidal we can assume the displacement starts at 0
- Wave B only reaches a displacement of 0 at a phase angle of 100
- The wavelength of a wave corresponds to a phase difference of 360° , as both waves have the same frequency, they have the same wavelength too
- So if wave B has a phase angle of 100° at a displacement of 0:
 - $\text{Phase} = 100 - 360 = -260^\circ$
- Wave A reaches a displacement of 0 and a phase angle of 0°
- So, the phase difference between wave A and B = $0 - (-260) = 260^\circ$

3

The correct answer is **A** because:

- The wave formed along the rope is a transverse wave
- A transverse wave travels at 90° to the direction of travel, this is to the right as shown in the diagram
- The particles can only oscillate in a vertical motion (up and down)
- The diagram below shows the wave in an instant later, moving the wave to the right



- The particle Q will move according to the form of the wave, a sudden jump up, followed by a high value of displacement and then with a final negative peak
- The answer therefore is **A**

B is incorrect because the graph shows a long negative displacement with a positive peak, this is the opposite direction to what would happen to particle Q

C is incorrect because the graph shows a positive peak with a long negative displacement

D is incorrect because the graph shows a negative peak then long positive displacement which is the opposite order to what would happen to particle Q

4

The correct answer is **C** because:

- Frequency $f = 15 \text{ Hz}$
- Speed $v = 24 \text{ m s}^{-1}$
 - The wave equation is $v = f \lambda$
- Wavelength $\lambda = \frac{v}{f} = \frac{24}{15} = 1.6 \text{ m}$
 - $1.6 \text{ m} = 1 \text{ cycle of the wave}$
- The question states that there is a phase difference of 2 m so $\frac{2}{1.6} = 1 \frac{1}{4}$
- A phase difference that is equal to $1 \frac{1}{4}$ is the same as saying it has a phase difference of $\frac{1}{4}$

A is incorrect because the distance between two points would be 2.4 m if they were out of phase by half a cycle

B is incorrect because if there was no phase difference then the wavelength and the distance between two points would be equal

D is incorrect because the distance between two points would be 3.0 m if they were out of phase by 0.8 of a cycle

5

The correct answer is **D** because:

- The intensity of light I is proportional to the square of the amplitude A

- $A^2 \propto I$
- The amplitude of the emerging light, let's call this $A_{em} = A \cos 60^\circ$
- Intensity of the emerging light is proportional to $A_{em}^2 = A^2 \cos^2(60^\circ) = 0.250 A^2$
 - $A_{em}^2 \propto I$
 - $0.250 A^2 \propto I$
- Therefore, the intensity of the emerging light = $0.250 I$

6

The correct answer is **D** because:

- The Doppler effect is when the sound heard by an observer changes depending on the relative motion between the source and the observer
- In this scenario, the source is the police car, and the person is the observer
- When the source moves towards the observer the waves bunch together resulting in the wavelength decreasing, so, frequency increases from the original emitted value
- When the source moves away from the observer the waves spread out resulting in the wavelength increasing, so, the frequency decreases from the original emitted value
- In this situation the Doppler effect is described by the following equation
 - $f_o = f_s \frac{(v + v_o)}{(v + v_s)}$
- From the question
 - v_o is 0 as they are stationary
 - v_f is 30 m s^{-1}
 - v is 340 m s^{-1}
 - f_s is 440 Hz
- $f_o = 440 \frac{(340 + 0)}{(340 + 30)} = 404.32 \text{ Hz}$

7

- $A_{em}^2 \propto I$
- $0.250 A^2 \propto I$

- Therefore, the intensity of the emerging light = 0.250 /

6

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 - $f_o = f_s \frac{(v + v_o)}{(v + v_s)}$
- From the question
 - v_o is 0 as they are stationary
 - v_f is 30 m s^{-1}
 - v is 340 m s^{-1}
 - f_s is 440 Hz
- $f_o = 440 \frac{(340 + 0)}{(340 + 30)} = 404.32 \text{ Hz}$

7

The correct answer is **B** because:

- The graph shows displacement against time, where the displacement is given in metres and the time in milliseconds
- It is possible to calculate the period from the graph since there are 3.5 waves in 12.2 seconds
 - $3.5 T = 12.2 \times 10^{-3} \text{ s}$

- Period, $T = \frac{12.2 \times 10^{-3}}{3.5} = 0.0035 \text{ s}$
- Frequency, $f = \frac{1}{T} = \frac{1}{0.0035} = 290 \text{ Hz}$

- The amplitude is the maximum displacement of the wave, this can be read from the y axis = 6.5 mm = 0.0065 m

8

The correct answer is **A** because :

- Intensity is directly proportional to the square of the amplitude, i.e.
 - $I \propto A^2 \text{ or } \frac{I_1}{I_2} = \frac{a_1^2}{a_2^2}$
- Where I is the intensity of the wave and a is the amplitude of the wave
- So, we get:
 - $\frac{2.5 \text{ W m}^{-2}}{I_2} = \frac{0.2 \text{ mm}^2}{0.4 \text{ mm}^2} = \frac{0.04}{0.16} = \frac{1}{4}$
 - $\therefore I_2 = \frac{4}{1} \times 2.5 \text{ W m}^{-2} = 10 \text{ W m}^{-2}$

9

The correct answer is **D** because:

- The time base setting is $5 \mu\text{s mm}^{-1}$
- From the diagram, each division is 2 mm so each division is $10 \mu\text{s}$
- 2.5 waves occupy 30 divisions
 - $2.5 T = 30 \times 10$
 - $T = \frac{30 \times 10}{2.5}$
- The period of the wave is $T = 120 \mu\text{s} = 12 \times 10^{-5} \text{ s}$
- Frequency = $\frac{1}{T} = \frac{1}{(12 \times 10^{-5})} = 8333.33 \text{ Hz} = 8.3 \text{ kHz}$

10

The correct answer is **B** because:

- Intensity I is proportional to amplitude squared A^2 , i.e.
 - $A^2 \propto I$ or $A \propto \sqrt{I}$
- If intensity is halved, $\frac{1}{2}$, then the new amplitude is proportional to $\sqrt{\frac{1}{2}}$
 - $A \propto \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}} \sqrt{1}$
- The new amplitude is therefore $\frac{1}{\sqrt{2}} A$
 - Therefore, the amplitude is reduced by a factor of $\frac{1}{\sqrt{2}} = 0.71$ which is shown in graph B

A is incorrect because the graph shows that the amplitude is halved this is not the case as it is reduced by a factor of 0.71

C & D are incorrect because the graphs show the same amplitude and increased frequency

11

The correct answer is A because:

- Intensity I is proportional to amplitude squared A^2
 - $I \propto A^2$
- From the equation given in the question, we can see that
 - $I \propto \frac{1}{r^2}$
- So, A^2 is inversely proportional to r^2 : $A^2 \propto \frac{1}{r^2}$
 - Therefore, $A \propto \frac{1}{r}$

